

An Overview of the Kähler Minimal Model Program

Kähler MMP

Yi Li

1 Overview

The aim of this note is to give a brief survey on recent developments in the Kähler minimal model program. This note is intended as an overview of what is currently known about the Kähler minimal model program, and briefly sketches some of the key ideas that appear along the way.

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2 Cone Theorems

The original proof of the cone theorem relies on Mori's bend-and-break technique. Unfortunately, this technique remains unavailable for Kähler manifolds. Using the recent resolution of the BDPP conjecture [Ou25], the cone theorem for Kähler varieties has nonetheless been established.

Theorem 2.1 ([HP24, Theorem 0.5]). *Let X be a compact $\mathbb{Q}$ -factorial Kähler variety of dimension n such that $(X, B + \beta)$ is generalized klt (resp. and $K_X + B + \beta_X$ is pseudo-effective). Then the following hold:*

(1) (Cone theorem) *There are at most countably many rational curves $\{\Gamma_i\}_{i \in I}$ such that*

$$\overline{\text{NA}}(X) = \overline{\text{NA}}(X)_{K_X + B + \beta_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i].$$

(2) (Length of extremal rays) $0 < -(K_X + B + \beta_X) \cdot \Gamma_i \leq 2n$.

(3) (Finiteness) *If $B + \beta_X$ or $K_X + B + \beta_X$ is big, then I is finite.*

💡 Idea of proof. A key observation is that the maximal-dimension component Z of the null locus $\text{Null}(\alpha)$ is covered by rational curves (to be proved below).

Let us give a remark on the history of cone theorems in the Kähler minimal model program: In [HP16], Höring–Petersen developed a Mori bend-and-break theorem for Kähler threefolds (to be discussed in the next section), using which they proved the cone theorem for Kähler threefolds with terminal singularities. Later, [CHP16, DO24] use a similar idea to generalise it to the cone theorem for Kähler threefold dlt pairs. Motivated by [CH20], [HP16] completes the proof of the cone theorem for Kähler klt pairs in any dimension.

3 Rational Curves on Kähler Varieties and Ou's BDPP Theorem for Compact Kähler Manifolds

The aim of this section is to give a brief survey of rational curves on the Kähler varieties (manifolds).

One of the breakthrough results is Ou's proof [Ou25] of the BDPP conjecture for compact Kähler manifolds.

Theorem 3.1 ([Ou25, Theorem 1]). *Let X be a compact Kähler manifold. Then K_X is not pseudo-effective if and only if X is uniruled.*

💡 **Idea of proof.**

When the singularities are bad (e.g. worse than klt), the BDPP theorem fails, as shown by the following example of Kollár.

Example 3.2.

Mori bend-and-break is a well-established theory for projective varieties.

4 Contraction Theorems

Currently, the contraction theorem is known only for Kähler threefolds. Höring–Peternell [HP16] proved it for $\mathbb{Q}$ -factorial compact Kähler threefolds with terminal singularities. Das–Hacon [DH25, DH24a] subsequently established the result in full for Kähler threefolds with $\mathbb{Q}$ -factorial dlt singularities.

Theorem 4.1 ([HP16, Theorem 1.3 + Corollary 7.9 + Theorem 7.12]). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at most terminal singularities such that K_X is pseudo-effective. Let $\mathbb{R}^+ [\Gamma_i]$ be a K_X -negative extremal ray in $\overline{\text{NA}}(X)$. Then the contraction of $\mathbb{R}^+ [\Gamma_i]$ exists in the Kähler category.*

(1) (Divisorial contraction when $n(\alpha) = 0$) Suppose that the extremal ray R is divisorial and $n(\alpha) = 0$. Then the surface S can be blown down to a point: there exists a bimeromorphic morphism $\varphi : X \rightarrow Y$ to a normal compact threefold Y with $\dim \varphi(S) = 0$ such that $\varphi|_{X \setminus S}$ is an isomorphism onto $Y \setminus \{p\}$, where $p = \varphi(S)$.

(2) (Divisorial contraction when $n(\alpha) = 1$) Suppose that the extremal ray R is divisorial and $n(\alpha) = 1$. Then there exists a fibration $f : S \rightarrow B$ onto a curve B such that a curve $C \subset S$ is contracted if and only if $[C] \in R$.

(3) (Flipping contraction case) Suppose that the extremal ray $R = \mathbb{R}^+ [\Gamma_i]$ is small, and let C be the locus covered by curves in R (cf. Notation 7.10). Then C is contractible.

💡 **Idea of proof.** We will not give proof details here, but let us sketch the key point of the proof.

Das–Hacon proved the contraction theorem for compact $\mathbb{Q}$ -factorial Kähler dlt pairs.

Theorem 4.2 ([DH24a, Theorem 1.2]). *Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler threefold dlt pair such that $K_X + B$ is pseudo-effective.*

Let ω be a Kähler class such that $\alpha = K_X + B + \omega$ is nef and big, and $\alpha^\perp \cap \overline{NA}(X) = R$ is an extremal ray.

Then there exists a projective bimeromorphic morphism $f : X \rightarrow Z$ with connected fibers such that $\alpha = f^ \alpha_Z$, where α_Z is a Kähler class on Z .*

*Moreover, if f is a divisorial contraction, then (Z, f_*B) has $\mathbb{Q}$ -factorial klt singularities.*

When a flipping contraction exists, Das–Hacon proved the following result on the existence of flips for generalized klt Kähler pairs in any dimension.

Theorem 4.3 ([DH24a, Theorem 5.12]). *Let $(X, B + \beta)$ be a $\mathbb{Q}$ -factorial gdlt pair, where X is a compact analytic variety belonging to Fujiki’s class C , and $f : X \rightarrow Z$ a flipping contraction. Then the flip of f exists.*

 **Idea of proof.**

5 Base Point Freeness

The transcendental base-point-freeness theorem remains wide open. For projective pairs, the base-point-free theorem holds.

Theorem 5.1 ([DH24b, Theorem 3.5]). *Let $(X, B + \beta)$ be a compact Kähler generalized klt pair such that $B + \beta$ is big. If $K_X + B + \beta_X$ is nef, then it is semi-ample.*

6 Existence of Minimal Models, Good Minimal Models, and Canonical Models

The existence of minimal models is completely known in the threefold case, and also for effective pairs on fourfolds. In the higher-dimensional setting, the existence of good minimal models has also been established under the maximal Albanese dimension hypothesis.

6.1 Existence of Minimal Models for Kähler Threefolds

Theorem 6.1 ([HP16, Theorem 1.1]). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at most terminal singularities. Suppose that K_X is pseudo-effective or, equivalently, that X is not uniruled. Then X has a minimal model, i.e. there exists an MMP*

$$X \rightarrow X'$$

such that $K_{X'}$ is nef.

Theorem 6.2 ([DH24a, Theorem 1.4]). *Let (X, B) be a dlt pair where X is a $\mathbb{Q}$ -factorial compact Kähler threefold. If $K_X + B$ is pseudo-effective, then there exists a finite sequence of flips and divisorial contractions*

$$\phi : X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_n$$

such that $K_{X_n} + \phi_* B$ is nef.

Later, Das–Hacon–Yáñez proved the result for generalized Kähler threefolds.

Theorem 6.3 ([DHY23, Theorem 1.2]). *Let $(X, B + \beta)$ be a generalized compact klt Kähler threefold pair.*

(1) *If $K_X + B + \beta_X$ is big, then $(X, B + \beta)$ has a log terminal model $f : X \dashrightarrow X^m$ and a unique log canonical model $g : X^m \rightarrow X^c$.*

(2) *If $K_X + B + \beta_X$ is pseudo-effective and β_X is big, then $K_X + B + \beta_X$ has a log terminal model $f : X \dashrightarrow X^m$, and there is a contraction $g : X^m \rightarrow Z$ such that $f_*(K_X + B + \beta_X) \equiv g^* \omega_Z$, where ω_Z is a Kähler form on Z .*

6.2 Existence of Minimal Models for Kähler Fourfold Effective Pairs

For effective Kähler fourfold dlt pairs, the existence of minimal models was proved by Das–Hacon–Păun.

Theorem 6.4 ([DHP24, Theorem 1.1]). *Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler fourfold dlt pair such that $K_X + B \sim_{\mathbb{Q}} M \geq 0$. Then (X, B) has a log minimal model.*

Theorem 6.5 ([DHP24, Corollary 7.12]). *Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler plt pair of dimension 4 such that $\kappa(X, K_X + B) \geq 0$. Then (X, B) has a log terminal model.*

6.3 Existence of Good Minimal Models for Kähler Varieties of Maximal Albanese Dimension

Das–Hacon also proved the existence of good minimal models for Kähler klt pairs of maximal Albanese dimension.

Theorem 6.6 ([DH24c, Theorem 1]). *Let (X, B) be a compact Kähler klt pair of maximal Albanese dimension. Then (X, B) has a good minimal model.*

💡 Idea of proof. (The proof idea here is only sketched; more details will be provided in the topic discussion notes.) We first take a log resolution and replace the Albanese map by its projective model $a' : X' \rightarrow A$, as illustrated in the diagram below.

$$\begin{array}{ccc} X' & \xrightarrow{\mu} & X \\ & \searrow a' & \downarrow a \\ & & A \end{array}$$

Note that the existence of a good minimal model is preserved under the modification $\mu : X' \rightarrow X$. We then divide the proof into two cases.

Case 1. When $\kappa(K_X + B) = 0$.

Let D be an irreducible component of the unique effective divisor

$$G \in |m(K_X + B)|.$$

We then show that the cohomological support locus satisfies

$$V^0(\omega_D) = \emptyset.$$

Using Fourier–Mukai techniques, we deduce that

$$a_*\omega_D = 0,$$

and therefore D is a -exceptional. We then run the relative MMP

$$\begin{array}{ccc} (X, B) & \overset{h}{\dashrightarrow} & (X', B') \\ & \searrow a & \swarrow a' \\ & & A \end{array}$$

with

$$K_{X'} + B' \sim_{\mathbb{Q}} E \geq 0,$$

where E is a' -exceptional (its exceptional support being the divisor identified above). By the negativity lemma, we can then deduce that $E = 0$, i.e.

$$K_{X'} + B' \sim_{\mathbb{Q}} 0.$$

In particular, (X, B) admits a good minimal model.

Case 2. When $\kappa(K_X + B) > 0$.

In this case, the proof is more involved. The idea is as follows. We first consider the Iitaka fibration

$$X \dashrightarrow Z,$$

whose very general fiber F satisfies the conditions of Case 1. Therefore, we can run the relative MMP for F over A , obtaining a complex torus F' . We then have the following commutative diagram (details on its construction will be given in the topic discussion notes):

$$\begin{array}{ccccc} X & & & & \\ \downarrow f & \searrow & a & \searrow & \\ & \bar{A} & \xrightarrow{\quad} & A & \\ & \downarrow & \lrcorner & \downarrow & \\ Z & \longrightarrow & \bar{Z} & \longrightarrow & A' = A/K \end{array}$$

Here, $\bar{A} = A \times_{A'} \bar{Z}$, the induced map $Z \rightarrow \bar{Z} = \text{Proj } R(K_X + B)$, and $K = a(F')$. We then decompose the log canonical divisor as

$$K_X + B \sim_{\mathbb{Q}} H_X + E,$$

where H_X is semi-ample (as the pullback of some ample divisor $\bar{H}$ on $\bar{Z}$) and $E \geq 0$ is an $\bar{A}$ -exceptional divisor. (Details on the construction of this decomposition will be provided in the topic discussion notes.)

Next, we run the $(K_X + B + (2n + 3)sH_X)$ -MMP over A :

$$\begin{array}{ccc} (X, B + (2n + 3)sH_X) & \dashrightarrow & (X', B' + (2n + 3)sH'_X) \\ & \searrow & \swarrow \\ & A & \end{array}$$

Using the bound on the length of extremal rays, we show that the relative MMP above is also an MMP over $\bar{Z}$, and hence over $\bar{A}$. Since $E \geq 0$ is $\bar{A}$ -exceptional, the negativity lemma implies $E' = 0$. Therefore,

$$K_{X'} + B' \sim_{\mathbb{Q}} \mu^* H_{\bar{A}},$$

for some semi-ample divisor $H_{\bar{A}}$ pulled back via $\mu : X' \rightarrow \bar{A}$.

Remark 6.7 (Comparison with the projective case). Fujino [Fuj13, Theorem 4.3] proved that projective klt pairs of maximal Albanese dimension admit good minimal models. It is instructive to compare Das–Hacon’s result with Fujino’s and identify what is new

in the Kähler setting.

6.4 Existence of Good Minimal Models When General Fibers Admit Good Minimal Models

Recently, Huang proved the existence of good minimal models for klt Kähler pairs when the Albanese map is projective and the general fibers admit good minimal models.

Theorem 6.8 ([Hua25, Theorem 1]). *Let $a_X : X \rightarrow A$ be the Albanese morphism of a compact Kähler variety X , where (X, B) is a klt pair. Assume a_X is projective with connected fibers, and let F be the general fiber of a_X . If (F, B_F) has a good minimal model, where $B_F = B|_F$ is the restricted boundary, then (X, B) has a good minimal model.*

💡 **Idea of proof.**

7 Projectivity Criterion for Kähler Varieties

Höring–Claudon developed a projectivity criterion for Kähler morphisms.

Theorem 7.1 ([CH24, Theorem 1.1]). *Let $f : X \rightarrow Y$ be a fibration between compact Kähler manifolds. Assume one of the following holds:*

- (1) *The natural map $f^* : H^0(Y, \Omega_Y^2) \rightarrow H^0(X, \Omega_X^2)$ is an isomorphism; or*
 - (2) *The morphism f is Moishezon, i.e. there exists a line bundle L on X that is f -big.*
- Then f is a projective morphism.*

💡 **Idea of proof.**

8 Canonical Bundle Formulas

The canonical bundle formula for proper morphisms between compact Kähler manifolds remains an open question. Some partial results are known for projective morphisms between compact Kähler varieties [HP24].

Theorem 8.1 ([HP24]). *[To be added.]*

9 Abundance Conjecture for Kähler Pairs

Abundance for lc pairs on Kähler threefolds was proved by Demailly–Ou [DO24].

Theorem 9.1 ([DO24, Theorem 1.1] + [DO25, Corollary 1.3]). *Let (X, Δ) be an lc pair such that X is a compact Kähler threefold. If $K_X + \Delta$ is nef, then it is semi-ample.*

 **Idea of proof.** Let us stress the differences between Kähler abundance and projective abundance.

10 Deformation of Positivity in Kähler Families

Hacon–Li–Rao proved the following deformation result for pseudo-effective adjoint classes in families.

Theorem 10.1 ([?]). *[To be added.]*

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